Houlin (Richard) Wang

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Research Interests

Phonetics and phonology, with a focus on prosodic structure and segmental variation in Beijing Mandarin; the interaction of tone, stress, and morphosyntax; computational methods for annotating and modeling casual and dialectal speech; and linguistically informed evaluation of automatic speech recognition for under-documented languages.

Education

Ph.D. in Linguistics Sep 2022 – Expected Jun 2027
University of California, Santa Cruz | Advisor: Ryan Bennett

Master of Arts in Linguistics Sep 2021 – Jun 2022
University of California, Los Angeles | Oral Exam Committee: Sun-Ah Jun (chair), Jesse Harris, Claire Moore-Cantwell | GPA 3.97

Bachelor of Arts in Linguistics, Magna Cum Laude Sep 2018 – Jun 2021
University of California, Los Angeles | Specialization in Computation; Minor in Japanese | GPA 3.95

Non-degree program Summer 2023
LSA Summer Institute, University of Massachusetts, Amherst

Publications

Proceedings

Wang, Richard. 2025. Distribution of tone affects prosodic parsing in Beijing Mandarin: Evidence from retroflex lenition. *Proceedings of the Annual Meetings on Phonology* 1(1). [link]

Wang, Richard. 2025. Morphosyntax-Prosody Mismatch in Beijing Mandarin: Evidence from Retroflex Lenition. *Proceedings of the 41st West Coast Conference on Formal Linguistics*, 592–599. [pdf]

Under Review

Wang, Richard et al. Under review. Does Omnilingual work? Testing ASR functionality on training languages and zero-shot contexts in a set of related languages. *The 10th Workshop on the Use of Computational Methods in the Study of Endangered Languages (ComputEL-10).*

Edited Volumes

Toosarvandani, Maziar, Nikolas Webster, Yağmur Kiper, Richard Wang & Larry Lyu (eds). *Proceedings of the 41st West Coast Conference on Formal Linguistics.*

Presentations

Posters

Wang, Richard. 2026. The Dual Nature of Retroflex Lenition: Categorical Prosodic Constraints and Gradient Phonetic Effects. Accepted, Hanyang International Symposium on Phonetics and Cognitive Sciences of Language. [abstract]

Wang, Richard. 2024. Distribution of neutral tone and retroflex lenition in Beijing Mandarin. 100th Annual Meeting of the Linguistic Society of America, New York, NY. Jan 2. [poster]

Wang, Richard. 2023. Distribution of neutral tone and retroflex lenition in Beijing Mandarin. Annual Meeting on Phonology, online, hosted by Johns Hopkins University. Oct 13. [poster]

Wang, Richard. 2023. Morphosyntactically conditioned footing in Beijing Mandarin: Evidence from

retroflex lenition. 41st West Coast Conference on Formal Linguistics, UC Santa Cruz, Santa Cruz, CA. May 4. [poster]

Invited Talks

Wang, Richard. 2024. Morphosyntactically conditioned footing in Beijing Mandarin: Evidence from retroflex lenition. Phorum Reading Group, UC Berkeley, Berkeley, CA. Oct 14.

Software

mandarin-wordlist-aligner: forced alignment of read word-list recordings (known-list scoring, MFA boundary repair) and an acoustic lenition classifier. Code only; participant data not included.

asr-error-metrics: word and character error rates that keep the alignment, with pooled corpus rates and substitution analysis.

Research Experience

Computational Linguistics Researcher Sep 2022 – Present
University of California, Santa Cruz

- Build Python and R pipelines for acoustic feature extraction, automated annotation, and quality control, analyzing speech at 10x the scale of manual methods
- Run mixed-effects regression and ML classification across 20,000+ model fits and 8 variables, showing that duration does not predict retroflex lenition in Beijing Mandarin
- Develop a lenition classifier on public Mandarin corpora (AISHELL-3, THCHS-30) for corpus quality monitoring
- Designed online speech-perception experiments (80+ participants) and modeled listener decision boundaries in R (Sep 2024 – Jun 2025)

AI Evaluation Research Fellow Jun 2026 – Present
Living Tongues Institute for Endangered Languages (remote)

- Evaluate Meta's Omnilingual ASR on 5 low-resource Munda languages (300M–7B models), scoring WER, CER, and phone matching against human IPA transcription
- Model 1,557 aligned Ho phone pairs with Random Forest and XGBoost (AUC $>$ 0.98): errors are structured by phonological features, mainly vowel height and backness
- Measure Ho phone error of 50.7% (strict), 42.7% (normalized), and 39.9% (broad), separating notational mismatch from recognition errors
- Diagnose decoder looping, wrong-script output, and mislabeled training data; annotate 90+ hours of speech audio and standardize quality control across a remote team

Research Mentor & Instructor Jun 2026 – Aug 2026
Science Internship Program, University of California, Santa Cruz

- Designed and taught an 8-week curriculum in Python, R, Praat, and hypothesis testing (10 hours/week, 80 hours total)
- Mentored 3 student researchers (10 hours/week) in annotating and modeling 2,500+ speech tokens

Research Assistant, UCLA Phonology Lab Apr 2020 – Jun 2021
Supervisor: Claire Moore-Cantwell

- Wrote Python code to analyze the gradience of Samoan thematic consonants under Symbolic Gradient Representation
- Built Python algorithms for phonotactic patterns across 10,000+ lexicon entries and reproducible R pipelines for corpus analysis

Research Assistant, UCLA Language Processing Lab Sep 2019 – Jun 2020
Supervisor: Jesse Harris

- Designed experimental materials, oversaw participants (30+), and collected data for a pupillometry experiment on cleft sentence parsing

Employment

MAE ToBI Annotator Summer 2020
DefinedCrowd Corporation

Teaching

Teaching Assistant, University of California, Santa Cruz

- Ling 80K: Invented Languages (Supervisor: Andrew Angeles) Spring 2026
- Ling 50: Introduction to Linguistics (Supervisor: Andrew Angeles) Winter 2026
- Ling 100: Phonetics I (Supervisor: Jaye Padgett) Fall 2025
- Ling 80K: Invented Languages (Supervisor: Andrew Angeles) Summer 2025
- Ling 151: Phonetics II (Instructor of Record: Jaye Padgett) Spring 2025
- Ling 80M: Language and Indigeneity in Mesoamerica (Instr. of Record: Ryan Bennett) Winter 2025
- Ling 102: Phonology II (Instructor: Rachel Walker) Fall 2024
- Ling 50: Introduction to Linguistics (Instructor: Amanda Rysling) Summer 2024
- Ling 50: Introduction to Linguistics (Instructor: Andrew Angeles) Spring 2024
- Ling 101: Phonology I (Instructor: Amanda Rysling) Winter 2024
- Ling 80M: Language and Indigeneity in Mesoamerica (Instructor: Ryan Bennett) Winter 2024
- Ling 100: Phonetics I (Instructor: Amanda Rysling) Fall 2023
- Ling 50: Introduction to Linguistics (Instructor: Matt Wagers) Summer 2023

Fellowships, Honors & Awards

- THI Summer Public Fellowship, UCSC Humanities Institute ($6,000) 2026
- THI Summer Dissertation Fellowship, UCSC Humanities Institute ($5,000) 2026
- Science Internship Program Fellowship, UCSC ($4,750) 2026
- Graduate Communications Fellowship, UCSC Humanities Institute ($1,400) 2023–2024
- LSA Linguistic Institute Fellowship, Linguistic Society of America ($2,500) 2023
- Department Scholar, UCLA Linguistics Department 2021–2022
- Departmental Honors, UCLA Linguistics Department ($250) 2021
- Helen R. Lane Merit Scholarship Award for Summer Research ($2,000) 2020

Service

- Member, Equity in Linguistics (EquiL), UCSC Linguistics Department 2023–2025
- Editor, What's Happening in Santa Cruz (WHASC) newsletter, UCSC Linguistics 2023–2024
- Entertainment Committee Chair, UCSC Linguistics Department 2023–2024

Skills

Phonetics & Corpus Methods: ASR evaluation (WER, CER), forced alignment (MFA), Praat, ELAN, corpus annotation, ToBI annotation
ML & Statistics: Mixed-effects regression (lme4), ML classification, bootstrapping, hypothesis testing
Programming: Python (pandas, NumPy, SciPy), R (tidyverse), SQL, PyTorch, Hugging Face, Git, Bash, C++, Haskell, LaTeX
Languages: Mandarin Chinese (native), English (fluent), German and Japanese (working proficiency)
Fieldwork and hardware: Setswana, Sierra Popoluca, Khalkha Mongolian; eyetracking, ultrasound

Last updated October 2026